```
import { useState, useRef, useEffect } from "react";

const SYS = {
  "Domestic Hot Water":   { m:"#c0392b", a:"DHW", t:"#fff", d:"125 PSI @ 210F", r
  "Domestic Cold Water":  { m:"#1a6fa8", a:"CW",  t:"#fff", d:"125 PSI @ 60F",  r
  "Steam":                { m:"#b7950b", a:"STM", t:"#000", d:"15 PSI @ 250F",  r
  "Condensate Return":    { m:"#0e6655", a:"CR",  t:"#fff", d:"30 PSI @ 212F",  r
  "Chilled Water Supply": { m:"#154360", a:"CWS", t:"#fff", d:"125 PSI @ 45F",  r
  "Chilled Water Return": { m:"#1a5276", a:"CWR", t:"#fff", d:"125 PSI @ 55F",  r
  "Hydronic Heating":     { m:"#943126", a:"HHW", t:"#fff", d:"125 PSI @ 180F", r
  "Process Piping":       { m:"#6c3483", a:"PRO", t:"#fff", d:"VARIES",         r
  "Gas (Med Pressure)":   { m:"#9a7d0a", a:"GAS", t:"#000", d:"5 PSI MAX",      r
  "Fire Protection":      { m:"#c0392b", a:"FP",  t:"#fff", d:"175 PSI",        r
};
const gs = s => SYS[s] || SYS["Domestic Hot Water"];

function buildJob(inp) {
  const ft = Math.max(parseFloat(inp.runLength) || 100, 10);
  const sz = parseFloat(inp.pipeSize) || 2;
  const hs = sz >= 6 ? 16 : sz >= 4 ? 14 : sz >= 2 ? 10 : 8;
  const nv = Math.max(2, Math.round(ft / 50));
  const isR = /floor|riser|vertical|rise|story/i.test(inp.jobDesc || "");
  const sc = gs(inp.systemType);
  const nFl = isR ? 3 : 0;
  const seg1 = isR ? ft * 0.36 : ft;
  const seg2 = isR ? ft * 0.22 : 0;
  const seg3 = isR ? ft * 0.42 : 0;
  const testPSI = Math.round((parseFloat(sc.d) || 125) * 1.5);
  const bh = sz <= 1 ? 0.18 : sz <= 2 ? 0.25 : sz <= 3 ? 0.32 : sz <= 4 ? 0.38 :
  const jf = inp.joiningMethod.includes("Weld") ? 1.55 : inp.joiningMethod === "G
  const bf = inp.buildingType === "Hospital / Healthcare" ? 1.3 : inp.buildingTyp
  const sf = /oshpd/i.test(inp.special || "") ? 1.35 : /confined/i.test(inp.speci
  const base = ft * bh * jf * bf * sf;
  const minH = Math.round(base * 0.85);
  const maxH = Math.round(base * 1.15);
  const pm = sc.ret ? 2.1 : 1.05;
  const mats = [
    { c:"PIPE",     i:`${inp.pipeSize} ${inp.material} Pipe`,            q:Math.r
    { c:"FITTINGS", i:`${inp.pipeSize} 90 deg Elbow`,                    q:Math.r
    { c:"FITTINGS", i:`${inp.pipeSize} 45 deg Elbow`,                    q:Math.r
    { c:"FITTINGS", i:`${inp.pipeSize} Straight Tee`,                    q:Math.r
    { c:"FITTINGS", i:`${inp.pipeSize} Reducing Tee`,                    q:Math.m
    { c:"FITTINGS", i:`${inp.pipeSize} Concentric Reducer`,              q:Math.m
    { c:"FITTINGS", i:`${inp.pipeSize} Union`,                           q:Math.m
```

```
    { c:"VALVES",   i:`${inp.pipeSize} Gate Valve OS&Y`,                 q:nv,
    { c:"VALVES",   i:`${inp.pipeSize} Globe Valve`,                     q:Math.m
    { c:"VALVES",   i:`${inp.pipeSize} Swing Check Valve`,               q:Math.m
    { c:"VALVES",   i:`${inp.pipeSize} Ball Valve Full Port`,            q:Math.m
    { c:"VALVES",   i:`${inp.pipeSize} Y-Strainer 1/16in Mesh`,         q:Math.ma
    { c:"INST",     i:"Pressure Gauge 0-160 PSI 2.5in Dial w/ Siphon",  q:Math.ma
    { c:"INST",     i:"Thermometer 9in Stem Adj Angle w/ Thermowell",   q:Math.ma
    { c:"INST",     i:"Auto Air Vent 1/2in FPT Float Type",             q:isR?nFl
    { c:"INST",     i:"Drain Valve 3/4in Gate Hose End",                q:Math.ma
    { c:"SUPPORTS", i:`${inp.pipeSize} Clevis Hanger ANVIL Fig.260`,    q:Math.ro
    { c:"SUPPORTS", i:"3/8in A307 All-Thread Rod x 10ft",               q:Math.ro
    { c:"SUPPORTS", i:`${inp.pipeSize} Riser Clamp ANVIL Fig.261`,      q:isR?nFl
    { c:"MISC",     i:`${inp.pipeSize} Expansion Loop Shop Fab`,         q:Math.m
    { c:"MISC",     i:"4x4 Galv Pipe Sleeve",                           q:isR?nFl
    { c:"MISC",     i:"UL Firestop Sealant 3M CP-25WB+",               q:isR?nFl:
  ];
  if (sc.ins) mats.push({ c:"INSUL", i:`${inp.pipeSize} Fiberglass Insulation 1in
  if (/seismic|oshpd/i.test(inp.special||"")) mats.push({ c:"SUPPORTS", i:"Seismi
  if (inp.joiningMethod.includes("Weld")) mats.push({ c:"MISC", i:"Welding Rod E7
  const dwgNo = "L-" + String(Math.floor((ft*7+sz*317)%900+100)).padStart(3,"0");
  const today = new Date().toLocaleDateString("en-US",{month:"2-digit",day:"2-dig
  return {
    routing: isR
      ? `Vertical ${inp.pipeSize} ${inp.material} ${inp.systemType.toLowerCase()}
      : `Horizontal ${inp.pipeSize} ${inp.material} pipe at EL.+${Math.round(8+sz
    compliance: `2022 CPC, OSHA 29 CFR 1926 Subpart P, CBC 2022 Ch.16, ASHRAE 90.
    laborNotes: `Mobilization and layout:  ${Math.round(minH*.08)}-${Math.round(m
    prefab: `Fabricate in ${Math.min(20,Math.round(ft/8))}-ft shop spools. Number
    mats, laborHours:`${minH}-${maxH}`, isR, hs, nv, sc, seg1, seg2, seg3, nFl, t
  };
}


// ══════════════════════════════════════════════════
// DRAWING ENGINE — ME STANDARD
// Line weights per ASME Y14.2:
//   Object/visible: 0.6mm (thick)
//   Hidden:         0.3mm
//   Dimension:      0.3mm
//   Centerline:     0.3mm
//   Text:           0.25mm min
// Canvas px approximation at 150dpi: 1mm ≈ 5.9px
// ══════════════════════════════════════════════════

const LW = {
THICK:   2.5,
MEDIUM:  1.5,
```

```
THIN:    0.8,
PHANTOM: 0.5,
ULTRA:   0.3,
};
function initCanvas(ctx, W, H) {
ctx.fillStyle = "#ffffff";
ctx.fillRect(0, 0, W, H);
ctx.fillStyle = "rgba(252,250,245,0.6)";
ctx.fillRect(0, 0, W, H);
// VERSION STAMP — confirms new code is running
ctx.fillStyle = "#c8860a";
ctx.fillRect(30, 30, W-60, 4);
ctx.fillRect(30, H-34, W-60, 4);
ctx.strokeStyle = "#000"; ctx.lineWidth = 4;
ctx.strokeRect(6, 6, W-12, H-12);
ctx.lineWidth = 1.2;
ctx.strokeRect(30, 30, W-60, H-60);
ctx.fillStyle = "#f0f0f0";
ctx.fillRect(6, 6, 24, H-12);
ctx.fillRect(W-30, 6, 24, H-12);
ctx.fillRect(6, 6, W-12, 24);
ctx.fillRect(6, H-30, W-12, 24);
ctx.strokeStyle = "#999"; ctx.lineWidth = 0.4;
const zw = (W-60)/6, zh = (H-60)/6;
for (let i=1; i<6; i++) {
ctx.beginPath(); ctx.moveTo(30+i*zw, 6); ctx.lineTo(30+i*zw, 30); ctx.stroke();
ctx.beginPath(); ctx.moveTo(30+i*zw, H-30); ctx.lineTo(30+i*zw, H-6); ctx.stroke(
ctx.beginPath(); ctx.moveTo(6, 30+i*zh); ctx.lineTo(30, 30+i*zh); ctx.stroke();
ctx.beginPath(); ctx.moveTo(W-30, 30+i*zh); ctx.lineTo(W-6, 30+i*zh); ctx.stroke(
}
ctx.fillStyle = "#333";
ctx.font = "bold 8px Arial,sans-serif";
ctx.textAlign = "center";
"ABCDEF".split("").forEach((l,i) => {
const x = 30+i*zw+zw/2;
ctx.fillText(l, x, 21);
ctx.fillText(l, x, H-8);
});
for (let i=0; i<6; i++) {
const y = 30+i*zh+zh/2;
ctx.fillText(String(i+1), 18, y+4);
ctx.fillText(String(i+1), W-18, y+4);
}
ctx.strokeStyle = "rgba(0,0,200,0.04)"; ctx.lineWidth = 0.3;
for (let x=30; x<=W-30; x+=15) {
ctx.beginPath(); ctx.moveTo(x, 30); ctx.lineTo(x, H-30); ctx.stroke();
}
```

```
for (let y=30; y<=H-30; y+=15) {
ctx.beginPath(); ctx.moveTo(30, y); ctx.lineTo(W-30, y); ctx.stroke();
}
ctx.strokeStyle = "rgba(0,0,200,0.09)"; ctx.lineWidth = 0.4;
for (let x=30; x<=W-30; x+=75) {
ctx.beginPath(); ctx.moveTo(x, 30); ctx.lineTo(x, H-30); ctx.stroke();
}
for (let y=30; y<=H-30; y+=75) {
ctx.beginPath(); ctx.moveTo(30, y); ctx.lineTo(W-30, y); ctx.stroke();
}
}
function titleBlock(ctx, W, H, inp, job, view, sheet) {
const bh = 82, by = H-bh-6, cw = W-60;
ctx.fillStyle = "#faf8f2";
ctx.fillRect(30, by, cw, bh);
ctx.strokeStyle = "#000"; ctx.lineWidth = 2;
ctx.strokeRect(30, by, cw, bh);
ctx.lineWidth = 1;
ctx.beginPath(); ctx.moveTo(30, by+bh*0.5); ctx.lineTo(30+cw, by+bh*0.5); ctx.str
const cols = [0.24, 0.47, 0.63, 0.79];
cols.forEach(c => {
ctx.beginPath(); ctx.moveTo(30+c*cw, by); ctx.lineTo(30+c*cw, by+bh); ctx.stroke(
});
const TX = (t, x, y, fs, col, bold, maxW) => {
ctx.fillStyle = col || "#000";
ctx.font = (bold ? "bold " : "") + fs + "px Arial,sans-serif";
ctx.textAlign = "left";
if (maxW) {
let s = String(t);
while (s.length > 2 && ctx.measureText(s).width > maxW) s = s.slice(0,-1);
if (s !== String(t)) s = s.slice(0,-2) + "..";
ctx.fillText(s, x, y);
} else {
ctx.fillText(String(t), x, y);
}
};
const p = 6, sc = job.sc;
const mw = (f) => cw*(cols[f]-(f>0?cols[f-1]:0))-p*2;
const c0=30+p, c1=30+cols[0]*cw+p, c2=30+cols[1]*cw+p, c3=30+cols[2]*cw+p, c4=30+
const r1=by+12, r2=by+26, r3=by+38, r4=by+bh*0.5+14, r5=by+bh*0.5+27;
TX("PROJECT", c0,r1, 6.5,"#666");
TX("UA LOCAL 38 — PIPE AI", c0,r2, 10,"#000",true, mw(0));
TX(inp.buildingType.toUpperCase(), c0,r3, 8,"#000",false, mw(0));
TX("SAN FRANCISCO, CA", c0,r4, 8.5,"#000",false, mw(0));
TX("SFDBI | UA LOCAL 38 | UA DISTRICT 16", c0,r5, 7,"#666",false, mw(0));
TX("SYSTEM / SPEC", c1,r1, 6.5,"#666");
ctx.fillStyle = sc.m; ctx.font = "bold 9.5px Arial,sans-serif"; ctx.textAlign = "
```

```
let sys = inp.systemType.toUpperCase(); while(sys.length>2 && ctx.measureText(sys
ctx.fillText(sys, c1, r2);
TX(inp.pipeSize+" "+inp.material.substring(0,18).toUpperCase(), c1,r3, 7.5,"#000"
TX(inp.joiningMethod.toUpperCase(), c1,r4, 8,"#000",false,mw(1));
TX("["+sc.a+"]  "+inp.runLength+" LF  |  "+sc.d, c1,r5, 7,"#666",false,mw(1));
TX("DRAWING TYPE", c2,r1, 6.5,"#666");
TX(view, c2,r2, 8.5,"#c8860a",true,mw(2));
TX("SYMBOLS: ASME Y32.2.3", c2,r3, 7,"#666",false,mw(2));
ctx.fillStyle="#c0392b"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="lef
ctx.fillText("NOT FOR CONSTRUCTION", c2,r4);
TX("VERIFY DIMS IN FIELD", c2,r5, 7,"#888",false,mw(2));
TX("DATE", c3,r1, 6.5,"#666");
TX(job.today, c3,r2, 9,"#000",true);
TX("DRAWN:  PIPE AI v8.0", c3,r3, 7,"#666");
TX("CHECKED:  ___________", c3,r4, 7,"#999");
TX("APPROVED: ___________", c3,r5, 7,"#999");
TX("DWG NO.", c4,r1, 6.5,"#666");
ctx.fillStyle="#000"; ctx.font="bold 16px 'Courier New',monospace"; ctx.textAlign
ctx.fillText(job.dwgNo, c4, r2+3);
TX("REV: A", c4,r3, 9,"#000",true);
TX("SHEET "+sheet+" OF 3", c4,r4, 9,"#000",true);
TX("TEST: "+job.testPSI+" PSI HYD.", c4,r5, 7.5,"#666");
}
function dim(ctx, x1, y1, x2, y2, lbl, side, col) {
side = side||1; col = col||"#000";
const dx=x2-x1, dy=y2-y1, len=Math.sqrt(dx*dx+dy*dy);
if (len < 4) return;
const nx=-dy/len, ny=dx/len, off=side*36;
ctx.strokeStyle=col; ctx.lineWidth=LW.THIN; ctx.setLineDash([]);
ctx.beginPath(); ctx.moveTo(x1+nx*4,y1+ny*4); ctx.lineTo(x1+nx*off,y1+ny*off); ct
ctx.beginPath(); ctx.moveTo(x2+nx*4,y2+ny*4); ctx.lineTo(x2+nx*off,y2+ny*off); ct
ctx.beginPath(); ctx.moveTo(x1+nx*off,y1+ny*off); ctx.lineTo(x2+nx*off,y2+ny*off)
const aL=9, aW=3;
[[x1,y1,1],[x2,y2,-1]].forEach(([px,py,d]) => {
const ax=dx/len*d, ay=dy/len*d;
ctx.fillStyle=col;
ctx.beginPath();
ctx.moveTo(px+nx*off, py+ny*off);
ctx.lineTo(px+nx*off+ax*aL-ny*aW, py+ny*off+ay*aL+nx*aW);
ctx.lineTo(px+nx*off+ax*aL+ny*aW, py+ny*off+ay*aL-nx*aW);
ctx.closePath(); ctx.fill();
});
const mx=(x1+x2)/2+nx*off, my=(y1+y2)/2+ny*off+(side>0?-7:16);
ctx.font = "bold 8.5px Arial,sans-serif";
const tw = ctx.measureText(lbl).width+10;
ctx.fillStyle="#fff"; ctx.fillRect(mx-tw/2,my-11,tw,14);
ctx.fillStyle=col; ctx.textAlign="center"; ctx.fillText(lbl,mx,my);
```

```
}
function leader(ctx, x1, y1, x2, y2, text, col, alignRight) {
col = col||"#000"; alignRight = alignRight!==false;
ctx.strokeStyle=col; ctx.lineWidth=LW.THIN;
ctx.fillStyle=col; ctx.beginPath(); ctx.arc(x1,y1,2.5,0,Math.PI*2); ctx.fill();
ctx.beginPath(); ctx.moveTo(x1,y1); ctx.lineTo(x2,y2); ctx.stroke();
const tail = alignRight ? 34 : -34;
ctx.beginPath(); ctx.moveTo(x2,y2); ctx.lineTo(x2+tail,y2); ctx.stroke();
ctx.font="7.5px Arial,sans-serif"; ctx.textAlign=alignRight?"left":"right";
ctx.fillStyle=col;
ctx.fillText(text, x2+tail+(alignRight?3:-3), y2+3);
}
function northArrow(ctx, x, y) {
ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
ctx.beginPath(); ctx.arc(x,y,20,0,Math.PI*2); ctx.stroke();
ctx.fillStyle="#000";
ctx.beginPath(); ctx.moveTo(x,y-18); ctx.lineTo(x+7,y+9); ctx.lineTo(x,y+5); ctx.
ctx.fillStyle="#fff";
ctx.beginPath(); ctx.moveTo(x,y-18); ctx.lineTo(x-7,y+9); ctx.lineTo(x,y+5); ctx.
ctx.strokeStyle="#000"; ctx.lineWidth=0.8; ctx.stroke();
ctx.lineWidth=0.5;
ctx.beginPath(); ctx.moveTo(x-20,y); ctx.lineTo(x+20,y); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x,y-20); ctx.lineTo(x,y+20); ctx.stroke();
ctx.fillStyle="#000"; ctx.font="bold 11px Arial,sans-serif"; ctx.textAlign="cente
ctx.fillText("N", x, y-22);
}
function tagBox(ctx, x, y, tag) {
if (!tag) return;
ctx.font = "bold 7.5px Arial,sans-serif";
const tw = ctx.measureText(tag).width+10;
ctx.fillStyle="#fffce0"; ctx.strokeStyle="#000"; ctx.lineWidth=0.8;
ctx.fillRect(x-tw/2, y, tw, 13);
ctx.strokeRect(x-tw/2, y, tw, 13);
ctx.fillStyle="#000"; ctx.textAlign="center";
ctx.fillText(tag, x, y+9.5);
}
function symGate(ctx, x, y, S, tag) {
ctx.fillStyle="#000"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.beginPath(); ctx.moveTo(x-S,y-S); ctx.lineTo(x-S,y+S); ctx.lineTo(x,y); ctx.c
ctx.beginPath(); ctx.moveTo(x+S,y-S); ctx.lineTo(x+S,y+S); ctx.lineTo(x,y); ctx.c
ctx.fillStyle="#444"; ctx.lineWidth=1;
ctx.fillRect(x-4,y-S-11,8,12); ctx.strokeRect(x-4,y-S-11,8,12);
ctx.lineWidth=1.5; ctx.strokeStyle="#000";
ctx.beginPath(); ctx.moveTo(x,y-S-11); ctx.lineTo(x,y-S-24); ctx.stroke();
ctx.lineWidth=1.5;
ctx.beginPath(); ctx.arc(x,y-S-31,11,0,Math.PI*2); ctx.stroke();
ctx.lineWidth=1;
```

```
[[0,11],[11,0],[0,-11],[-11,0]].forEach(([dx,dy])=>{
ctx.beginPath();ctx.moveTo(x,y-S-31);ctx.lineTo(x+dx,y-S-31+dy);ctx.stroke();
});
ctx.lineWidth=1.5;
ctx.beginPath();ctx.moveTo(x-11,y-S-42);ctx.lineTo(x+11,y-S-42);ctx.stroke();
tagBox(ctx, x, y+S+5, tag);
}
function symGlobe(ctx, x, y, S, tag) {
ctx.fillStyle="#000"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.beginPath(); ctx.moveTo(x-S,y-S); ctx.lineTo(x-S,y+S); ctx.lineTo(x,y); ctx.c
ctx.beginPath(); ctx.moveTo(x+S,y-S); ctx.lineTo(x+S,y+S); ctx.lineTo(x,y); ctx.c
ctx.fillStyle="#fff"; ctx.lineWidth=1.2;
ctx.beginPath(); ctx.arc(x,y,S*0.52,0,Math.PI*2); ctx.fill(); ctx.stroke();
ctx.fillStyle="#444"; ctx.lineWidth=1;
ctx.fillRect(x-4,y-S-11,8,12); ctx.strokeRect(x-4,y-S-11,8,12);
ctx.lineWidth=1.5; ctx.strokeStyle="#000";
ctx.beginPath(); ctx.moveTo(x,y-S-11); ctx.lineTo(x,y-S-23); ctx.stroke();
ctx.beginPath(); ctx.arc(x,y-S-30,9,0,Math.PI*2); ctx.stroke();
ctx.lineWidth=1;
ctx.beginPath();ctx.moveTo(x-9,y-S-30);ctx.lineTo(x+9,y-S-30);ctx.stroke();
tagBox(ctx, x, y+S+5, tag);
}
function symCheck(ctx, x, y, S, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.beginPath(); ctx.moveTo(x-S,y-S); ctx.lineTo(x-S,y+S); ctx.lineTo(x,y); ctx.c
ctx.fill(); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x,y-S); ctx.lineTo(x,y+S); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x,y); ctx.lineTo(x+S,y); ctx.stroke();
ctx.fillStyle="#000";
ctx.beginPath(); ctx.arc(x,y-S,3,0,Math.PI*2); ctx.fill();
tagBox(ctx, x, y+S+5, tag);
}
function symBall(ctx, x, y, S, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.fillRect(x-S,y-S,S*2,S*2); ctx.strokeRect(x-S,y-S,S*2,S*2);
ctx.lineWidth=1.2;
ctx.beginPath(); ctx.arc(x,y,S*0.52,0,Math.PI*2); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x-S*0.52,y); ctx.lineTo(x+S*0.52,y); ctx.stroke();
ctx.lineWidth=2;
ctx.beginPath(); ctx.moveTo(x,y-S); ctx.lineTo(x,y-S-16); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x-9,y-S-16); ctx.lineTo(x+9,y-S-16); ctx.stroke();
tagBox(ctx, x, y+S+5, tag);
}
function symButterfly(ctx, x, y, S, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.beginPath(); ctx.arc(x,y,S,0,Math.PI*2); ctx.fill(); ctx.stroke();
ctx.lineWidth=1.2;
```

```
ctx.beginPath(); ctx.moveTo(x-S,y); ctx.bezierCurveTo(x-S/2,y-S*0.75,x+S/2,y-S*0.
ctx.beginPath(); ctx.moveTo(x-S,y); ctx.bezierCurveTo(x-S/2,y+S*0.75,x+S/2,y+S*0.
ctx.lineWidth=1.5;
ctx.beginPath(); ctx.moveTo(x,y-S); ctx.lineTo(x,y-S-14); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x-8,y-S-14); ctx.lineTo(x+8,y-S-14); ctx.stroke();
tagBox(ctx, x, y+S+5, tag);
}
function symStrainer(ctx, x, y, S, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.THICK;
ctx.beginPath();
ctx.moveTo(x-S,y); ctx.lineTo(x,y-S); ctx.lineTo(x+S,y); ctx.lineTo(x,y+S);
ctx.closePath(); ctx.fill(); ctx.stroke();
ctx.lineWidth=1.2;
ctx.beginPath(); ctx.moveTo(x,y-S*0.48); ctx.lineTo(x,y+S*0.4); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x-S*0.48,y+S*0.1); ctx.lineTo(x,y+S*0.4); ctx.stroke(
ctx.beginPath(); ctx.moveTo(x+S*0.48,y+S*0.1); ctx.lineTo(x,y+S*0.4); ctx.stroke(
ctx.lineWidth=0.5;
for (let i=-2;i<=2;i++) {
ctx.beginPath();ctx.moveTo(x+i*S*0.2,y-S*0.28);ctx.lineTo(x+i*S*0.2,y+S*0.32);ctx
}
tagBox(ctx, x, y+S+5, tag);
}
function symGauge(ctx, x, y, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
ctx.beginPath(); ctx.arc(x,y,12,0,Math.PI*2); ctx.fill(); ctx.stroke();
ctx.lineWidth=0.8;
for (let a=0; a<5; a++) {
const ang = -Math.PI*0.8 + a*Math.PI*0.4;
ctx.beginPath();
ctx.moveTo(x+Math.cos(ang)*8, y+Math.sin(ang)*8);
ctx.lineTo(x+Math.cos(ang)*12, y+Math.sin(ang)*12);
ctx.stroke();
}
ctx.lineWidth=1; ctx.strokeStyle="#000";
ctx.beginPath(); ctx.moveTo(x,y); ctx.lineTo(x-4,y-8); ctx.stroke();
ctx.fillStyle="#000"; ctx.font="bold 7px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText("P",x+4,y+4);
ctx.lineWidth=1.2;
ctx.beginPath(); ctx.moveTo(x,y+12); ctx.lineTo(x,y+18); ctx.stroke();
ctx.beginPath(); ctx.arc(x+4,y+22,4,Math.PI,0); ctx.stroke();
if (tag) {
ctx.fillStyle="#555"; ctx.font="7px Arial,sans-serif"; ctx.textAlign="center";
ctx.fillText(tag,x,y+33);
}
}
function symThermo(ctx, x, y, tag) {
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
```

```
ctx.fillRect(x-6,y-12,12,24); ctx.strokeRect(x-6,y-12,12,24);
ctx.fillStyle="#c0392b"; ctx.lineWidth=1;
ctx.beginPath(); ctx.arc(x,y+14,5.5,0,Math.PI*2); ctx.fill(); ctx.stroke();
ctx.fillStyle="#c0392b";
ctx.fillRect(x-2,y-8,4,22);
ctx.strokeStyle="#000"; ctx.lineWidth=0.6;
for (let i=0;i<5;i++) {
ctx.beginPath();ctx.moveTo(x+2,y-9+i*6);ctx.lineTo(x+6,y-9+i*6);ctx.stroke();
}
ctx.lineWidth=1.5; ctx.strokeStyle="#000";
ctx.beginPath(); ctx.moveTo(x,y+20); ctx.lineTo(x,y+28); ctx.stroke();
if (tag) {
ctx.fillStyle="#555"; ctx.font="7px Arial,sans-serif"; ctx.textAlign="center";
ctx.fillText(tag,x,y+38);
}
}
function symAV(ctx, x, y) {
ctx.fillStyle="#000"; ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
ctx.beginPath(); ctx.moveTo(x,y-13); ctx.lineTo(x-8,y+4); ctx.lineTo(x+8,y+4); ct
ctx.beginPath(); ctx.arc(x,y-17,4,0,Math.PI*2); ctx.fill();
ctx.lineWidth=1;
ctx.beginPath(); ctx.moveTo(x,y+4); ctx.lineTo(x,y+14); ctx.stroke();
ctx.fillStyle="#333"; ctx.font="7px Arial,sans-serif"; ctx.textAlign="center";
ctx.fillText("A/V",x,y+24);
}
function symDrain(ctx, x, y) {
ctx.fillStyle="#000"; ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
ctx.beginPath(); ctx.moveTo(x-7,y); ctx.lineTo(x+7,y); ctx.lineTo(x,y+13); ctx.cl
ctx.lineWidth=1;
ctx.beginPath(); ctx.moveTo(x,y); ctx.lineTo(x,y-15); ctx.stroke();
ctx.fillStyle="#333"; ctx.font="7px Arial,sans-serif"; ctx.textAlign="center";
ctx.fillText("D/V",x,y+24);
}
function symExpLoop(ctx, x, y, pR) {
const lh=54, lw=45;
ctx.strokeStyle="#000"; ctx.lineWidth=pR*2; ctx.lineCap="round";
ctx.beginPath();
ctx.moveTo(x,y);
ctx.lineTo(x,y-lh);
ctx.lineTo(x-lw,y-lh);
ctx.lineTo(x-lw,y+lh);
ctx.lineTo(x+lw,y+lh);
ctx.lineTo(x+lw,y-lh);
ctx.lineTo(x,y-lh);
ctx.lineTo(x,y);
ctx.stroke();
ctx.lineCap="butt";
```

```
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=0.8;
ctx.fillRect(x-30,y-lh-20,60,13); ctx.strokeRect(x-30,y-lh-20,60,13);
ctx.fillStyle="#000"; ctx.font="bold 7.5px Arial,sans-serif"; ctx.textAlign="cent
ctx.fillText("EXP. LOOP",x,y-lh-11);
}
function symReducer(ctx, x, y, pR) {
const rL=pR+5, rS=Math.max(3,pR*0.5);
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=1.5;
ctx.beginPath();
ctx.moveTo(x-13,y-rL); ctx.lineTo(x+13,y-rS);
ctx.lineTo(x+13,y+rS); ctx.lineTo(x-13,y+rL);
ctx.closePath(); ctx.fill(); ctx.stroke();
}
function symFlange(ctx, x, y, pR, dir) {
ctx.strokeStyle="#000"; ctx.lineWidth=2.8;
ctx.beginPath(); ctx.moveTo(x,y-pR-5); ctx.lineTo(x,y+pR+5); ctx.stroke();
ctx.beginPath(); ctx.moveTo(x+dir*7,y-pR-5); ctx.lineTo(x+dir*7,y+pR+5); ctx.stro
ctx.fillStyle="#000";
[y-pR, y, y+pR].forEach(by => {
ctx.beginPath(); ctx.arc(x+dir*7,by,2,0,Math.PI*2); ctx.fill();
});
ctx.lineWidth=1;
ctx.beginPath();ctx.moveTo(x+dir*12,y-pR-2);ctx.lineTo(x+dir*12,y+pR+2);ctx.strok
}
function symHanger(ctx, hx, strY, pipeY, pR) {
ctx.strokeStyle="#555"; ctx.lineWidth=1; ctx.setLineDash([3,2]);
ctx.beginPath(); ctx.moveTo(hx,strY); ctx.lineTo(hx,pipeY-pR-2); ctx.stroke();
ctx.setLineDash([]);
ctx.strokeStyle="#333"; ctx.lineWidth=2;
ctx.beginPath(); ctx.moveTo(hx-14,pipeY-pR-20); ctx.lineTo(hx+14,pipeY-pR-20); ct
ctx.beginPath(); ctx.moveTo(hx-14,pipeY-pR-20); ctx.lineTo(hx-14,pipeY-pR-8); ctx
ctx.beginPath(); ctx.moveTo(hx+14,pipeY-pR-20); ctx.lineTo(hx+14,pipeY-pR-8); ctx
ctx.fillStyle="#333";
ctx.beginPath(); ctx.arc(hx,strY,3,0,Math.PI*2); ctx.fill();
}
function renderPlan(canvas, job, inp) {
const ctx = canvas.getContext("2d");
const W = canvas.width, H = canvas.height;
initCanvas(ctx,W,H);
const ft  = Math.max(parseFloat(inp.runLength)||100,10);
const sz  = parseFloat(inp.pipeSize)||2;
const pR  = Math.min(10,Math.max(4,sz*1.75));
const sc  = job.sc;
const L=50,R=W-50,T=55,B=H-110;
const supY = T+(B-T)*(sc.ret?0.33:0.50);
const retY = T+(B-T)*0.67;
const strY = supY - pR - 100;
```

```
const mg   = L+130;
const endX = R-130;
const pxPerFt = (endX-mg)/ft;
const S   = Math.max(13,pR*2.0);
const minGap = Math.max(100,ft*pxPerFt/12);
ctx.fillStyle="rgba(80,90,110,0.06)";
for(let bx=mg;bx<endX;bx+=70) {
ctx.fillRect(bx,strY-28,62,28);
ctx.strokeStyle="rgba(60,70,90,0.18)"; ctx.lineWidth=0.6;
for(let hx=bx+6;hx<bx+60;hx+=9){ctx.beginPath();ctx.moveTo(hx,strY-28);ctx.lineTo
}
ctx.strokeStyle="rgba(40,50,80,0.55)"; ctx.lineWidth=3; ctx.setLineDash([]);
ctx.beginPath(); ctx.moveTo(mg-50,strY); ctx.lineTo(endX+50,strY); ctx.stroke();
ctx.fillStyle="rgba(40,50,80,0.6)"; ctx.font="8px Arial,sans-serif"; ctx.textAlig
ctx.fillText("STRUCTURAL STEEL FRAMING — SEE STRUCTURAL DRAWINGS",(mg+endX)/2,str
const drawPipe = (py, colorMain, label, flowRight) => {
if (sc.ins) {
const iW = pR*2.0;
ctx.fillStyle=`rgba(200,160,40,0.06)`;
ctx.fillRect(mg,py-iW/2,endX-mg,iW);
ctx.strokeStyle="rgba(160,110,30,0.5)"; ctx.lineWidth=0.8; ctx.setLineDash([6,4])
ctx.beginPath();ctx.moveTo(mg,py-iW/2);ctx.lineTo(endX,py-iW/2);ctx.stroke();
ctx.beginPath();ctx.moveTo(mg,py+iW/2);ctx.lineTo(endX,py+iW/2);ctx.stroke();
ctx.setLineDash([]);
const lix=mg+ft*pxPerFt*0.18;
leader(ctx,lix,py-iW/2,lix-20,py-iW/2-28,"INSUL. 1\" FBG/ASJ (TYP.)","#666",false
}
for(let f=job.hs;f<ft-4;f+=job.hs) {
symHanger(ctx,mg+f*pxPerFt,strY,py,pR);
}
ctx.strokeStyle="#000"; ctx.lineWidth=pR*2; ctx.lineCap="butt";
ctx.beginPath(); ctx.moveTo(mg,py); ctx.lineTo(endX,py); ctx.stroke();
const bw = Math.min(64,ft*pxPerFt*0.10);
const bx = mg+ft*pxPerFt*0.36;
ctx.fillStyle=colorMain; ctx.fillRect(bx,py-pR,bw,pR*2);
ctx.strokeStyle="#000"; ctx.lineWidth=0.6; ctx.strokeRect(bx,py-pR,bw,pR*2);
ctx.fillStyle=sc.t; ctx.font=`bold ${Math.max(9,pR*1.05)}px Arial,sans-serif`;
ctx.textAlign="center"; ctx.fillText(sc.a,bx+bw/2,py+pR*0.38);
ctx.strokeStyle="#000"; ctx.lineWidth=LW.PHANTOM; ctx.setLineDash([16,4,4,4]);
ctx.beginPath(); ctx.moveTo(mg-60,py); ctx.lineTo(endX+60,py); ctx.stroke();
ctx.setLineDash([]);
ctx.fillStyle="#000"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="left";
ctx.fillText("CL",mg-58,py-3);
ctx.fillStyle=colorMain; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="lef
ctx.fillText(label, mg, py-pR-30);
const midX=(mg+endX)/2;
ctx.fillStyle="#fff"; ctx.fillRect(midX-40,py-8,80,17);
```

```
ctx.strokeStyle="#000"; ctx.lineWidth=0.6; ctx.strokeRect(midX-40,py-8,80,17);
ctx.fillStyle="#000"; ctx.font="bold 8px 'Courier New',monospace"; ctx.textAlign=
ctx.fillText(inp.pipeSize+" "+inp.material.substring(0,14),midX,py+4);
const arStep=ft*pxPerFt/4;
const arDir = flowRight ? 1 : -1;
const arStart = flowRight ? mg+arStep*0.6 : endX-arStep*0.6;
ctx.fillStyle="rgba(0,0,0,0.7)";
const ax=arStart;
if(flowRight){
ctx.beginPath();ctx.moveTo(ax+13,py);ctx.lineTo(ax-1,py-6);ctx.lineTo(ax+2,py);ct
ctx.fillStyle="#000";ctx.font="7px Arial,sans-serif";ctx.textAlign="left";ctx.fil
} else {
ctx.beginPath();ctx.moveTo(ax-13,py);ctx.lineTo(ax+1,py-6);ctx.lineTo(ax-2,py);ct
ctx.fillStyle="#000";ctx.font="7px Arial,sans-serif";ctx.textAlign="right";ctx.fi
}
};
drawPipe(supY, sc.m, "SUPPLY — "+inp.pipeSize+" "+inp.material+" ["+sc.a+"]  "+sc
let lastX = -9999;
const supSyms = [
[0.04, x=>symStrainer(ctx,x,supY,S,"STR-1")],
[0.11, x=>symGauge(ctx,x,supY-pR-30,"PG-1")],
[0.17, x=>symThermo(ctx,x,supY-pR-26,"TI-1")],
[0.24, x=>symCheck(ctx,x,supY,S,"CV-1")],
[0.32, x=>{ ctx.fillStyle="#000";ctx.beginPath();ctx.arc(x,supY,pR*0.85,0,Math.PI
[0.40, x=>symGate(ctx,x,supY,S,"GV-1")],
[0.50, x=>symExpLoop(ctx,x,supY,pR)],
[0.59, x=>symBall(ctx,x,supY,S,"BV-1")],
[0.67, x=>symButterfly(ctx,x,supY,S,"BFV-1")],
[0.75, x=>symGlobe(ctx,x,supY,S,"GLV-1")],
[0.82, x=>symReducer(ctx,x,supY,pR)],
[0.88, x=>symGauge(ctx,x,supY-pR-30,"PG-2")],
[0.93, x=>symThermo(ctx,x,supY-pR-26,"TI-2")],
[0.97, x=>symAV(ctx,x,supY-pR-10)],
];
supSyms.forEach(([f,fn]) => {
const sx = mg+f*ft*pxPerFt;
if (sx>mg+20 && sx<endX-20 && sx-lastX>=minGap) { fn(sx); lastX=sx; }
});
symDrain(ctx,endX-22,supY+pR+8);
symFlange(ctx,mg,supY,pR,1);
ctx.fillStyle="#1a3060"; ctx.strokeStyle="#2471a3"; ctx.lineWidth=1.5;
ctx.fillRect(mg-96,supY-18,90,36); ctx.strokeRect(mg-96,supY-18,90,36);
ctx.fillStyle="#74b9ff"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("SUPPLY",mg-51,supY-3); ctx.fillText("SOURCE",mg-51,supY+10);
symFlange(ctx,endX,supY,pR,-1);
ctx.fillStyle="#601a1a"; ctx.strokeStyle="#c0392b"; ctx.lineWidth=1.5;
ctx.fillRect(endX+6,supY-18,90,36); ctx.strokeRect(endX+6,supY-18,90,36);
```

```
ctx.fillStyle="#fab1a0"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("SUPPLY",endX+51,supY-3); ctx.fillText("DEST.",endX+51,supY+10);
if (sc.ret) {
drawPipe(retY, sc.m, "RETURN — "+inp.pipeSize+" "+inp.material+" ["+sc.a+"]", fal
let lastRx=-9999;
const retSyms = [
[0.07, x=>symGate(ctx,x,retY,S,"GV-3")],
[0.28, x=>symStrainer(ctx,x,retY,S,"STR-2")],
[0.50, x=>symCheck(ctx,x,retY,S,"CV-2")],
[0.72, x=>symGate(ctx,x,retY,S,"GV-4")],
[0.87, x=>symGauge(ctx,x,retY-pR-30,"PG-3")],
[0.93, x=>symThermo(ctx,x,retY-pR-26,"TI-3")],
];
retSyms.forEach(([f,fn]) => {
const rx=mg+f*ft*pxPerFt;
if(rx>mg+20&&rx<endX-20&&rx-lastRx>=minGap){fn(rx);lastRx=rx;}
});
symDrain(ctx,endX-22,retY+pR+8);
symFlange(ctx,mg,retY,pR,1);
ctx.fillStyle="#1a3060"; ctx.strokeStyle="#2471a3"; ctx.lineWidth=1.5;
ctx.fillRect(mg-96,retY-18,90,36); ctx.strokeRect(mg-96,retY-18,90,36);
ctx.fillStyle="#74b9ff"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("RETURN",mg-51,retY-3); ctx.fillText("DEST.",mg-51,retY+10);
symFlange(ctx,endX,retY,pR,-1);
ctx.fillStyle="#601a1a"; ctx.strokeStyle="#c0392b"; ctx.lineWidth=1.5;
ctx.fillRect(endX+6,retY-18,90,36); ctx.strokeRect(endX+6,retY-18,90,36);
ctx.fillStyle="#fab1a0"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("RETURN",endX+51,retY-3); ctx.fillText("SOURCE",endX+51,retY+10);
dim(ctx,endX+80,supY,endX+80,retY,
Math.round((retY-supY)/pxPerFt*0.5+8)+"\"",
1,"#555");
}
dim(ctx,mg,supY+pR+14,endX,supY+pR+14,ft+"'-0\"  TOTAL RUN",-1,"#000");
if(ft>20&&pxPerFt>3){
const hx1=mg+job.hs*pxPerFt,hx2=mg+job.hs*2*pxPerFt;
if(hx2<endX-60) dim(ctx,hx1,supY-pR-14,hx2,supY-pR-14,job.hs+"'-0\" TYP.",1,"#555
}
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(mg-120,supY-16,88,32); ctx.strokeRect(mg-120,supY-16,88,32);
ctx.fillStyle="#000"; ctx.font="bold 7.5px Arial,sans-serif"; ctx.textAlign="cent
ctx.fillText("EL. +"+Math.round(8+sz*0.5)+"'-0\" AFF",mg-76,supY-2);
ctx.font="7px Arial,sans-serif"; ctx.fillText("SLOPE: "+( sz>=2?"1/8":"1/4" )+"\"
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(mg,T+40,268,80); ctx.strokeRect(mg,T+40,268,80);
ctx.fillStyle="#c8860a"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="lef
ctx.fillText("PIPE SPECIFICATION",mg+8,T+55);
ctx.strokeStyle="#000"; ctx.lineWidth=0.5;
```

```
ctx.beginPath();ctx.moveTo(mg,T+59);ctx.lineTo(mg+268,T+59);ctx.stroke();
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif";
[
"SIZE:    "+inp.pipeSize,
"MATERIAL: "+inp.material,
"SERVICE:  "+inp.systemType.substring(0,26),
"JOINING:  "+inp.joiningMethod,
"DESIGN:   "+sc.d,
"TEST:     "+job.testPSI+" PSI HYDROSTATIC",
].forEach((t,i)=>ctx.fillText(t,mg+8,T+71+i*11.5));
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(L+36,B-158,430,148); ctx.strokeRect(L+36,B-158,430,148);
ctx.fillStyle="#c8860a"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="lef
ctx.fillText("GENERAL NOTES:",L+44,B-144);
ctx.strokeStyle="#000"; ctx.lineWidth=0.5;
ctx.beginPath();ctx.moveTo(L+36,B-140);ctx.lineTo(L+466,B-140);ctx.stroke();
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif";
[
"1.  ALL WORK SHALL CONFORM TO: 2022 CPC, OSHA 29 CFR 1926, CBC 2022, AND ASHRAE
"2.  ALL PIPING SYMBOLS SHALL BE PER ASME Y32.2.3.",
"3.  HYDROSTATIC TEST: "+job.testPSI+" PSI FOR 4 HOURS, WITNESSED. INSPECTOR TO S
"4.  INSULATE ALL HOT AND CHILLED PIPING PER TITLE 24 PART 6.",
"5.  SEISMIC BRACING PER SMACNA IEMP 2ND EDITION AT ALL PRESCRIBED INTERVALS.",
"6.  HANGER SCHEDULE: ANVIL/GRINNELL/HILTI — ENGINEER-STAMPED REQUIRED.",
"7.  COORDINATE ALL TRADES PRIOR TO ROUGH-IN. MAINTAIN MINIMUM CLEARANCES.",
"8.  VERIFY ALL DIMENSIONS IN FIELD PRIOR TO SHOP FABRICATION.",
"9.  HANGER SPACING: "+job.hs+"'-0\" O.C. MAX — ANVIL FIG.260 CLEVIS HANGER.",
"10. SUBMIT SPOOL DRAWINGS TO GC A MINIMUM OF THREE (3) WEEKS BEFORE INSTALLATION
].forEach((n,i)=>ctx.fillText(n,L+44,B-128+i*11));
const rvX=R-190,rvY=B-158;
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(rvX,rvY,178,90); ctx.strokeRect(rvX,rvY,178,90);
ctx.fillStyle="#000"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText("REVISION RECORD",rvX+89,rvY+13);
ctx.lineWidth=0.5;
[[0,rvY+17,178,rvY+17],[24,rvY+17,24,rvY+90],[116,rvY+17,116,rvY+90]]
.forEach(([x1,y1,x2,y2])=>{ctx.beginPath();ctx.moveTo(rvX+x1,y1);ctx.lineTo(rvX+x
ctx.font="7px Arial,sans-serif"; ctx.textAlign="left"; ctx.fillStyle="#777";
ctx.fillText("REV",rvX+5,rvY+27); ctx.fillText("DATE",rvX+28,rvY+27); ctx.fillTex
ctx.fillStyle="#000";
ctx.fillText("A",rvX+7,rvY+41); ctx.fillText(job.today,rvX+28,rvY+41); ctx.fillTe
["B","C","D"].forEach((r,i)=>{
ctx.fillText(r,rvX+7,rvY+55+i*13);
ctx.fillStyle="#ccc"; ctx.fillText("______________",rvX+28,rvY+55+i*13); ctx.fill
});
northArrow(ctx,R-56,T+56);
const lx=R-196,ly=T+96,lw=184;
```

```
const litems=[
{d:(x,y)=>{ctx.fillStyle="#000";ctx.fillRect(x,y-4,22,9);},l:"PIPE — "+sc.a+" SYS
{d:(x,y)=>{ctx.strokeStyle="#000";ctx.lineWidth=LW.PHANTOM;ctx.setLineDash([12,3,
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symGate(ctx,0,0,
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symGlobe(ctx,0,0
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symCheck(ctx,0,0
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symBall(ctx,0,0,
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symButterfly(ctx
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symStrainer(ctx,
{d:(x,y)=>{symGauge(ctx,x+10,y,null);},l:"PRESSURE GAUGE (PG)"},
{d:(x,y)=>{symThermo(ctx,x+10,y,null);},l:"THERMOMETER (TI)"},
{d:(x,y)=>{symAV(ctx,x+10,y);},l:"AUTO AIR VENT (A/V)"},
{d:(x,y)=>{symDrain(ctx,x+10,y);},l:"DRAIN VALVE (D/V)"},
{d:(x,y)=>{ctx.strokeStyle="rgba(160,110,30,0.55)";ctx.lineWidth=5;ctx.setLineDas
{d:(x,y)=>{symFlange(ctx,x+11,y,5,1);},l:"FLANGED CONNECTION"},
];
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(lx,ly,lw,litems.length*14+30); ctx.strokeRect(lx,ly,lw,litems.length
ctx.fillStyle="#000"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="left";
ctx.fillText("LEGEND — ASME Y32.2.3",lx+6,ly+14);
ctx.lineWidth=0.5; ctx.beginPath();ctx.moveTo(lx,ly+18);ctx.lineTo(lx+lw,ly+18);c
litems.forEach((it,i) => {
const iy=ly+31+i*14;
ctx.save();ctx.fillStyle="#000";ctx.strokeStyle="#000";ctx.lineWidth=1.2;ctx.font
it.d(lx+4,iy); ctx.restore();
ctx.fillStyle="#111"; ctx.font="8px Arial,sans-serif"; ctx.textAlign="left"; ctx.
});
titleBlock(ctx,W,H,inp,job,"PLAN VIEW — SUPPLY & RETURN PIPING","1");
}
function renderIso(canvas, job, inp) {
const ctx = canvas.getContext("2d");
const W = canvas.width, H = canvas.height;
initCanvas(ctx,W,H);
const ft  = Math.max(parseFloat(inp.runLength)||100,10);
const sz  = parseFloat(inp.pipeSize)||2;
const pR  = Math.max(5,Math.min(11,sz*2.0));
const sc  = job.sc;
const isR = job.isR;
const c30 = Math.cos(Math.PI/6), s30 = Math.sin(Math.PI/6);
const L=50,R=W-50,T=60,B=H-120;
const nFl = job.nFl||3;
const {seg1,seg2,seg3} = job;
const rScl = 110;
const kpts = isR ? [
[0,0,0],[seg1,0,0],[seg1,0,seg2],[seg1,nFl,seg2],[seg1+seg3,nFl,seg2],
...[...Array(nFl+1)].flatMap((_,fl)=>[[0,fl,0],[(seg1+seg3)*0.94,fl,seg2*1.35]])
] : [[0,0,0],[ft,0,0]];
```

```
const raw1 = kpts.map(([wx,wy,wz])=>({sx:wx*c30-wz*c30,sy:-wy*rScl+wx*s30+wz*s30}
const xs=raw1.map(p=>p.sx),ys=raw1.map(p=>p.sy);
const minX=Math.min(...xs),maxX=Math.max(...xs);
const minY=Math.min(...ys),maxY=Math.max(...ys);
const rawW=maxX-minX||1, rawH=maxY-minY||1;
const padL=160,padR=210,padT=90,padB=110;
const availW=(R-L)-padL-padR, availH=(B-T)-padT-padB;
const scl = Math.min(availW/rawW, availH/rawH, 9.0);
const cx=L+padL+availW/2, cy=T+padT+availH/2;
const ox=cx-(minX+maxX)/2*scl, oy=cy-(minY+maxY)/2*scl;
const iso=(wx,wy,wz)=>({
sx: ox+wx*scl*c30-wz*scl*c30,
sy: oy-wy*rScl+wx*scl*s30+wz*scl*s30
});
const pLine=(p1,p2,specLines)=>{
const dx=p2.sx-p1.sx,dy=p2.sy-p1.sy,len=Math.sqrt(dx*dx+dy*dy);
if(len<2)return;
ctx.strokeStyle="rgba(0,0,0,0.08)"; ctx.lineWidth=pR*2+8; ctx.lineCap="round";
ctx.beginPath();ctx.moveTo(p1.sx+3,p1.sy+4);ctx.lineTo(p2.sx+3,p2.sy+4);ctx.strok
ctx.strokeStyle="#000"; ctx.lineWidth=pR*2; ctx.lineCap="round";
ctx.beginPath();ctx.moveTo(p1.sx,p1.sy);ctx.lineTo(p2.sx,p2.sy);ctx.stroke();
ctx.strokeStyle="#000"; ctx.lineWidth=LW.PHANTOM; ctx.setLineDash([14,4,4,4]);
ctx.beginPath();ctx.moveTo(p1.sx,p1.sy);ctx.lineTo(p2.sx,p2.sy);ctx.stroke();
ctx.setLineDash([]);
if(len>50){
const bL=Math.min(32,len*0.15),mx=(p1.sx+p2.sx)/2,my=(p1.sy+p2.sy)/2;
const bu=dx/len,bv=dy/len,bny=-bv*pR,bnx=bu*pR;
ctx.fillStyle=sc.m;
ctx.beginPath();
ctx.moveTo(mx-bu*bL+bny,my-bv*bL+bnx);ctx.lineTo(mx+bu*bL+bny,my+bv*bL+bnx);
ctx.lineTo(mx+bu*bL-bny,my+bv*bL-bnx);ctx.lineTo(mx-bu*bL-bny,my-bv*bL-bnx);
ctx.closePath();ctx.fill();
ctx.fillStyle=sc.t; ctx.font=`bold ${Math.max(7,pR*0.85)}px Arial,sans-serif`;
ctx.textAlign="center"; ctx.fillText(sc.a,mx,my+3);
}
if(specLines){
const lx=(p1.sx+p2.sx)/2, ly=(p1.sy+p2.sy)/2-pR-68;
let mw=0;
ctx.font="8px Arial,sans-serif";
specLines.forEach(s=>{const w=ctx.measureText(s).width+16;if(w>mw)mw=w;});
const lh=specLines.length*13+10;
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(lx-mw/2,ly,mw,lh); ctx.strokeRect(lx-mw/2,ly,mw,lh);
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif"; ctx.textAlign="left";
specLines.forEach((s,i)=>ctx.fillText(s,lx-mw/2+8,ly+12+i*13));
ctx.strokeStyle="#000"; ctx.lineWidth=0.8;
ctx.beginPath();ctx.moveTo(lx,ly+lh);ctx.lineTo(lx,(p1.sy+p2.sy)/2-pR-2);ctx.stro
```

```
ctx.fillStyle="#000"; ctx.beginPath();ctx.arc(lx,(p1.sy+p2.sy)/2-pR-2,2.5,0,Math.
}
};
const isoElbow=pt=>{
ctx.fillStyle="#000"; ctx.beginPath();ctx.arc(pt.sx,pt.sy,pR+4,0,Math.PI*2);ctx.f
ctx.strokeStyle="#c8860a"; ctx.lineWidth=2;
ctx.beginPath();ctx.arc(pt.sx,pt.sy,pR+7,0,Math.PI*2);ctx.stroke();
ctx.fillStyle="rgba(255,255,255,0.3)";
ctx.beginPath();ctx.arc(pt.sx-2,pt.sy-2,4,0,Math.PI*2);ctx.fill();
};
const isoGate=(pt,dx,dy,tag)=>{
const S=pR*2.2,nx=-dy*S,ny=dx*S;
ctx.fillStyle="#000"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.MEDIUM;
ctx.beginPath();ctx.moveTo(pt.sx-dx*S,pt.sy-dy*S);ctx.lineTo(pt.sx-dx*S+nx,pt.sy-
ctx.beginPath();ctx.moveTo(pt.sx+dx*S,pt.sy+dy*S);ctx.lineTo(pt.sx+dx*S+nx,pt.sy+
ctx.lineWidth=1.5;
ctx.beginPath();ctx.moveTo(pt.sx,pt.sy);ctx.lineTo(pt.sx+nx*1.8,pt.sy+ny*1.8);ctx
ctx.beginPath();ctx.arc(pt.sx+nx*1.8,pt.sy+ny*1.8,S*0.78,0,Math.PI*2);ctx.stroke(
ctx.lineWidth=1;
ctx.beginPath();ctx.moveTo(pt.sx+nx*1.8-S*0.78,pt.sy+ny*1.8);ctx.lineTo(pt.sx+nx*
if(tag){
ctx.fillStyle="#000"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText(tag,pt.sx+nx*3.4,pt.sy+ny*3.4+3);
}
};
const isoStrain=(pt,tag)=>{
const S=pR*1.8;
ctx.fillStyle="#fff"; ctx.strokeStyle="#000"; ctx.lineWidth=LW.MEDIUM;
ctx.beginPath();ctx.moveTo(pt.sx-S,pt.sy);ctx.lineTo(pt.sx,pt.sy-S);ctx.lineTo(pt
ctx.lineWidth=0.7;
for(let i=-2;i<=2;i++){ctx.beginPath();ctx.moveTo(pt.sx+i*S*0.22,pt.sy-S*0.32);ct
if(tag){ctx.fillStyle="#000";ctx.font="bold 8px Arial,sans-serif";ctx.textAlign="
};
const isoHanger=pt=>{
ctx.strokeStyle="#555"; ctx.lineWidth=1; ctx.setLineDash([3,2]);
ctx.beginPath();ctx.moveTo(pt.sx,pt.sy-pR-2);ctx.lineTo(pt.sx,pt.sy-pR-36);ctx.st
ctx.setLineDash([]);
ctx.strokeStyle="#333"; ctx.lineWidth=2;
ctx.beginPath();ctx.moveTo(pt.sx-14,pt.sy-pR-36);ctx.lineTo(pt.sx+14,pt.sy-pR-36)
ctx.beginPath();ctx.moveTo(pt.sx-14,pt.sy-pR-36);ctx.lineTo(pt.sx-14,pt.sy-pR-23)
ctx.beginPath();ctx.moveTo(pt.sx+14,pt.sy-pR-36);ctx.lineTo(pt.sx+14,pt.sy-pR-23)
ctx.fillStyle="#333"; ctx.beginPath();ctx.arc(pt.sx,pt.sy-pR-36,3,0,Math.PI*2);ct
};
const isoClamp=pt=>{
ctx.strokeStyle="#000"; ctx.lineWidth=3;
ctx.beginPath();ctx.moveTo(pt.sx-20,pt.sy-7);ctx.lineTo(pt.sx+20,pt.sy-7);ctx.str
ctx.beginPath();ctx.moveTo(pt.sx-20,pt.sy+7);ctx.lineTo(pt.sx+20,pt.sy+7);ctx.str
```

```
ctx.lineWidth=4;
ctx.beginPath();ctx.moveTo(pt.sx+20,pt.sy-16);ctx.lineTo(pt.sx+20,pt.sy+16);ctx.s
ctx.strokeStyle="#999"; ctx.lineWidth=0.7;
for(let i=-12;i<=12;i+=4){ctx.beginPath();ctx.moveTo(pt.sx+20,pt.sy+i);ctx.lineTo
ctx.fillStyle="#000";
[pt.sx-10,pt.sx+10].forEach(bx=>{ctx.beginPath();ctx.arc(bx,pt.sy,3,0,Math.PI*2);
ctx.fillStyle="#333"; ctx.font="bold 7px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText("RC",pt.sx,pt.sy+24);
};
const floorSlab=fl=>{
const fw=(seg1+seg3)*0.94, fd=seg2*1.35;
const pts=[iso(0,fl,0),iso(fw,fl,0),iso(fw,fl,fd),iso(0,fl,fd)];
ctx.fillStyle=fl%2===0?"rgba(145,165,205,0.2)":"rgba(125,145,185,0.14)";
ctx.beginPath();ctx.moveTo(pts[0].sx,pts[0].sy);pts.forEach(p=>ctx.lineTo(p.sx,p.
ctx.strokeStyle="rgba(30,45,90,0.6)"; ctx.lineWidth=2;
ctx.beginPath();ctx.moveTo(pts[0].sx,pts[0].sy);pts.forEach(p=>ctx.lineTo(p.sx,p.
ctx.strokeStyle="rgba(50,70,120,0.16)"; ctx.lineWidth=0.6;
for(let t2=0;t2<=1;t2+=0.05){
const x1=pts[0].sx+(pts[1].sx-pts[0].sx)*t2,y1=pts[0].sy+(pts[1].sy-pts[0].sy)*t2
ctx.beginPath();ctx.moveTo(x1,y1);ctx.lineTo(x1-12,y1+14);ctx.stroke();
}
const lp=iso(-0.07,fl,0);
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(lp.sx-86,lp.sy-12,82,17);ctx.strokeRect(lp.sx-86,lp.sy-12,82,17);
ctx.fillStyle="#000"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText("FL-"+(fl+1)+"  EL. +"+fl*14+"'-0\"",lp.sx-45,lp.sy+1);
ctx.fillStyle="#666"; ctx.font="7px Arial,sans-serif";
ctx.fillText("8\" CONC. SLAB",lp.sx-45,lp.sy+12);
};
const isoDim=(p1,p2,lbl)=>{
const dx=p2.sx-p1.sx,dy=p2.sy-p1.sy,l=Math.sqrt(dx*dx+dy*dy);
if(l<8)return;
const nx=-dy/l,ny=dx/l,off=30;
ctx.strokeStyle="#000"; ctx.lineWidth=LW.THIN;
ctx.beginPath();ctx.moveTo(p1.sx,p1.sy);ctx.lineTo(p1.sx+nx*off,p1.sy+ny*off);ctx
ctx.beginPath();ctx.moveTo(p2.sx,p2.sy);ctx.lineTo(p2.sx+nx*off,p2.sy+ny*off);ctx
ctx.beginPath();ctx.moveTo(p1.sx+nx*off,p1.sy+ny*off);ctx.lineTo(p2.sx+nx*off,p2.
const aL=9,aW=3;
[[p1,1],[p2,-1]].forEach(([p,d])=>{
ctx.fillStyle="#000"; ctx.beginPath();
ctx.moveTo(p.sx+nx*off,p.sy+ny*off);
ctx.lineTo(p.sx+nx*off+dx/l*d*aL-ny*aW,p.sy+ny*off+dy/l*d*aL+nx*aW);
ctx.lineTo(p.sx+nx*off+dx/l*d*aL+ny*aW,p.sy+ny*off+dy/l*d*aL-nx*aW);
ctx.closePath();ctx.fill();
});
const mx=(p1.sx+p2.sx)/2+nx*off,my=(p1.sy+p2.sy)/2+ny*off-7;
ctx.font="bold 8.5px Arial,sans-serif";
```

```
const tw=ctx.measureText(lbl).width+12;
ctx.fillStyle="#fff"; ctx.fillRect(mx-tw/2,my-11,tw,15);
ctx.fillStyle="#000"; ctx.textAlign="center"; ctx.fillText(lbl,mx,my);
};
if(isR) for(let fl=0;fl<=nFl;fl++) floorSlab(fl);
const s1s=iso(0,0,0), s1e=iso(seg1,0,0);
pLine(s1s,s1e,[inp.pipeSize+" "+inp.material,inp.joiningMethod+" — ["+sc.a+"]",sc
for(let f=job.hs;f<seg1;f+=job.hs) isoHanger(iso(f,0,0));
const isoGapMin=Math.max(110,(s1e.sx-s1s.sx)/5);
let lastIsoX=-9999;
[
[0.07,pt=>isoStrain(pt,"STR-1")],
[0.30,pt=>{const p2=iso(0.35*seg1,0,0);const ddx=p2.sx-pt.sx,ddy=p2.sy-pt.sy,ddl=
[0.60,pt=>{const p2=iso(0.65*seg1,0,0);const ddx=p2.sx-pt.sx,ddy=p2.sy-pt.sy,ddl=
[0.85,pt=>isoStrain(pt,"STR-2")],
].forEach(([f,fn])=>{
const pt=iso(f*seg1,0,0);
if(pt.sx-lastIsoX>=isoGapMin){fn(pt);lastIsoX=pt.sx;}
});
symAV(ctx,iso(seg1*0.93,0,0).sx,iso(seg1*0.93,0,0).sy-pR-9);
symDrain(ctx,iso(seg1*0.06,0,0).sx,iso(seg1*0.06,0,0).sy+pR+6);
if(isR){
isoElbow(s1e);
const s2e=iso(seg1,0,seg2);
pLine(s1e,s2e,null);
const dMid=iso(seg1,0,seg2*0.5);
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=0.9;
ctx.fillRect(dMid.sx+14,dMid.sy-10,130,14); ctx.strokeRect(dMid.sx+14,dMid.sy-10,
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif"; ctx.textAlign="left";
ctx.fillText(inp.pipeSize+" OFFSET — ["+sc.a+"]",dMid.sx+18,dMid.sy+1);
isoElbow(s2e);
const s3e=iso(seg1,nFl,seg2);
pLine(s2e,s3e,null);
const rMid=iso(seg1,nFl*0.5,seg2);
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=0.9;
ctx.fillRect(rMid.sx+14,rMid.sy-10,160,14); ctx.strokeRect(rMid.sx+14,rMid.sy-10,
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif"; ctx.textAlign="left";
ctx.fillText(inp.pipeSize+" RISER [VERT.] — "+sc.d,rMid.sx+18,rMid.sy+1);
for(let fli=1;fli<=nFl;fli++) isoClamp(iso(seg1,fli,seg2));
isoElbow(s3e);
const s4e=iso(seg1+seg3,nFl,seg2);
pLine(s3e,s4e,null);
for(let f2=job.hs;f2<seg3;f2+=job.hs) isoHanger(iso(seg1+f2,nFl,seg2));
const t1=iso(seg1+seg3*0.4,nFl,seg2),t1n=iso(seg1+seg3*0.45,nFl,seg2);
const t1dx=t1n.sx-t1.sx,t1dy=t1n.sy-t1.sy,t1dl=Math.sqrt(t1dx*t1dx+t1dy*t1dy);
isoGate(t1,t1dl>0?t1dx/t1dl:1,t1dl>0?t1dy/t1dl:0,"GV-3");
symAV(ctx,iso(seg1+seg3*0.87,nFl,seg2).sx,iso(seg1+seg3*0.87,nFl,seg2).sy-pR-9);
```

```
const dp=s4e;
ctx.fillStyle="#601a1a"; ctx.strokeStyle="#c0392b"; ctx.lineWidth=2;
ctx.beginPath();ctx.arc(dp.sx,dp.sy,pR+8,0,Math.PI*2);ctx.fill();ctx.stroke();
ctx.strokeStyle="#ff6b6b"; ctx.lineWidth=1;
ctx.beginPath();ctx.arc(dp.sx,dp.sy,pR+12,0,Math.PI*2);ctx.stroke();
ctx.fillStyle="#fab1a0"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("DEST.",dp.sx,dp.sy+pR+25);
isoDim(iso(0,0,0),iso(seg1,0,0),Math.round(seg1)+"'-0\"");
const rv1=iso(seg1,0,seg2),rv2=iso(seg1,nFl,seg2);
ctx.strokeStyle="#000"; ctx.lineWidth=LW.THIN; ctx.setLineDash([3,3]);
ctx.beginPath();ctx.moveTo(rv1.sx+36,rv1.sy);ctx.lineTo(rv2.sx+36,rv2.sy);ctx.str
ctx.setLineDash([]);
ctx.beginPath();ctx.moveTo(rv1.sx,rv1.sy);ctx.lineTo(rv1.sx+36,rv1.sy);ctx.stroke
ctx.beginPath();ctx.moveTo(rv2.sx,rv2.sy);ctx.lineTo(rv2.sx+36,rv2.sy);ctx.stroke
const rLbl=nFl+" FL — "+nFl*14+"'-0\" RISE";
ctx.font="bold 8.5px Arial,sans-serif";
const rtw=ctx.measureText(rLbl).width+14;
ctx.fillStyle="#fff"; ctx.fillRect(rv1.sx+38,(rv1.sy+rv2.sy)/2-11,rtw,15);
ctx.fillStyle="#000"; ctx.textAlign="left"; ctx.fillText(rLbl,rv1.sx+42,(rv1.sy+r
isoDim(iso(seg1,nFl,seg2),iso(seg1+seg3,nFl,seg2),Math.round(seg3)+"'-0\"");
} else {
const ep=iso(ft,0,0);
ctx.fillStyle="#601a1a"; ctx.strokeStyle="#c0392b"; ctx.lineWidth=2;
ctx.beginPath();ctx.arc(ep.sx,ep.sy,pR+8,0,Math.PI*2);ctx.fill();ctx.stroke();
ctx.strokeStyle="#ff6b6b"; ctx.lineWidth=1;
ctx.beginPath();ctx.arc(ep.sx,ep.sy,pR+12,0,Math.PI*2);ctx.stroke();
ctx.fillStyle="#fab1a0"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("DEST.",ep.sx,ep.sy+pR+25);
isoDim(iso(0,0,0),iso(ft,0,0),ft+"'-0\"  TOTAL RUN");
}
const sp=iso(0,0,0);
ctx.fillStyle="#1a3060"; ctx.strokeStyle="#2471a3"; ctx.lineWidth=2;
ctx.beginPath();ctx.arc(sp.sx,sp.sy,pR+8,0,Math.PI*2);ctx.fill();ctx.stroke();
ctx.strokeStyle="#74b9ff"; ctx.lineWidth=1;
ctx.beginPath();ctx.arc(sp.sx,sp.sy,pR+12,0,Math.PI*2);ctx.stroke();
ctx.fillStyle="#74b9ff"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="cen
ctx.fillText("SOURCE",sp.sx,sp.sy+pR+25);
const fa=iso(seg1*0.46,0,0),fa2=iso(seg1*0.52,0,0);
const fdx=fa2.sx-fa.sx,fdy=fa2.sy-fa.sy,fdl=Math.sqrt(fdx*fdx+fdy*fdy);
if(fdl>0){
ctx.fillStyle="rgba(255,255,255,0.9)";
ctx.beginPath();
ctx.moveTo(fa.sx+fdx/fdl*14,fa.sy+fdy/fdl*14);
ctx.lineTo(fa.sx-fdx/fdl*2+fdy/fdl*6,fa.sy-fdy/fdl*2-fdx/fdl*6);
ctx.lineTo(fa.sx,fa.sy);
ctx.lineTo(fa.sx-fdx/fdl*2-fdy/fdl*6,fa.sy-fdy/fdl*2+fdx/fdl*6);
ctx.closePath();ctx.fill();
```

```
ctx.strokeStyle="#000"; ctx.lineWidth=0.8; ctx.stroke();
}
const axO={sx:L+72,sy:B-56},axL=50;
[
[sc.a+" (E)",sc.m,{sx:axO.sx+axL*c30,sy:axO.sy-axL*s30}],
["Z (N)","#27ae60",{sx:axO.sx-axL*c30,sy:axO.sy-axL*s30}],
["Y (UP)","#2471a3",{sx:axO.sx,sy:axO.sy-axL}],
].forEach(([lbl,col,tp])=>{
ctx.strokeStyle=col; ctx.lineWidth=2.5;
ctx.beginPath();ctx.moveTo(axO.sx,axO.sy);ctx.lineTo(tp.sx,tp.sy);ctx.stroke();
const ddx=tp.sx-axO.sx,ddy=tp.sy-axO.sy,dl=Math.sqrt(ddx*ddx+ddy*ddy);
ctx.fillStyle=col;
ctx.beginPath();ctx.moveTo(tp.sx,tp.sy);
ctx.lineTo(tp.sx-ddx/dl*10+ddy/dl*3.5,tp.sy-ddy/dl*10-ddx/dl*3.5);
ctx.lineTo(tp.sx-ddx/dl*10-ddy/dl*3.5,tp.sy-ddy/dl*10+ddx/dl*3.5);
ctx.closePath();ctx.fill();
ctx.font="bold 8.5px Arial,sans-serif"; ctx.textAlign="center";
ctx.fillText(lbl,tp.sx+(tp.sx>axO.sx?18:tp.sx<axO.sx?-18:0),tp.sy+(tp.sy<axO.sy?-
});
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(L+36,B-90,438,78); ctx.strokeRect(L+36,B-90,438,78);
ctx.fillStyle="#c8860a"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="lef
ctx.fillText("NOTES:",L+44,B-76);
ctx.lineWidth=0.5; ctx.beginPath();ctx.moveTo(L+36,B-72);ctx.lineTo(L+474,B-72);c
ctx.fillStyle="#000"; ctx.font="8px Arial,sans-serif";
[
"1. ISOMETRIC IS DIAGRAMMATIC — NOT TO SCALE.",
"2. ALL SYMBOLS PER ASME Y32.2.3 — SINGLE-LINE PIPE CONVENTION.",
"3. VERIFY ALL DIMENSIONS AND ELEVATIONS IN FIELD BEFORE FABRICATION.",
"4. ALL PIPE SIZES NOMINAL UNLESS DIMENSIONED OTHERWISE.",
"5. WELD SYMBOLS PER AWS A2.4. SEE PLAN VIEW (SHEET 1) FOR COMPLETE NOTES.",
].forEach((n,i)=>ctx.fillText(n,L+44,B-58+i*11));
const rvX=R-190,rvY=B-90;
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(rvX,rvY,178,78); ctx.strokeRect(rvX,rvY,178,78);
ctx.fillStyle="#000"; ctx.font="bold 8px Arial,sans-serif"; ctx.textAlign="center
ctx.fillText("REVISION RECORD",rvX+89,rvY+13);
ctx.lineWidth=0.5;
[[0,rvY+17,178,rvY+17],[24,rvY+17,24,rvY+78],[116,rvY+17,116,rvY+78]]
.forEach(([x1,y1,x2,y2])=>{ctx.beginPath();ctx.moveTo(rvX+x1,y1);ctx.lineTo(rvX+x
ctx.font="7px Arial,sans-serif"; ctx.textAlign="left"; ctx.fillStyle="#777";
ctx.fillText("REV",rvX+5,rvY+27); ctx.fillText("DATE",rvX+28,rvY+27); ctx.fillTex
ctx.fillStyle="#000";
ctx.fillText("A",rvX+7,rvY+41); ctx.fillText(job.today,rvX+28,rvY+41); ctx.fillTe
["B","C","D"].forEach((r,i)=>{
ctx.fillText(r,rvX+7,rvY+55+i*13);
ctx.fillStyle="#ccc"; ctx.fillText("______________",rvX+28,rvY+55+i*13); ctx.fill
```

```
});
const lx=R-196,ly=T+54,lw=184;
const isoLgnd=[
{d:(x,y)=>{ctx.fillStyle="#000";ctx.fillRect(x,y-4,22,9);},l:"PIPE — "+sc.a+" SYS
{d:(x,y)=>{ctx.strokeStyle="#000";ctx.lineWidth=LW.PHANTOM;ctx.setLineDash([12,3,
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symGate(ctx,0,0,
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symCheck(ctx,0,0
{d:(x,y)=>{ctx.save();ctx.translate(x+11,y);ctx.scale(0.52,0.52);symStrainer(ctx,
{d:(x,y)=>{ctx.fillStyle="#000";ctx.beginPath();ctx.arc(x+11,y,5,0,Math.PI*2);ctx
{d:(x,y)=>{symAV(ctx,x+11,y);},l:"AUTO AIR VENT (A/V)"},
{d:(x,y)=>{symDrain(ctx,x+11,y);},l:"DRAIN VALVE (D/V)"},
{d:(x,y)=>{ctx.strokeStyle="#444";ctx.lineWidth=2;ctx.beginPath();ctx.moveTo(x,y-
];
if(isR){
isoLgnd.push({d:(x,y)=>{ctx.strokeStyle="#000";ctx.lineWidth=3;ctx.beginPath();ct
isoLgnd.push({d:(x,y)=>{ctx.fillStyle="rgba(145,165,205,0.5)";ctx.strokeStyle="#0
}
isoLgnd.push({d:(x,y)=>{ctx.fillStyle="#1a3060";ctx.strokeStyle="#2471a3";ctx.lin
const lh=isoLgnd.length*14+30;
ctx.fillStyle="#fffff8"; ctx.strokeStyle="#000"; ctx.lineWidth=1;
ctx.fillRect(lx,ly,lw,lh); ctx.strokeRect(lx,ly,lw,lh);
ctx.fillStyle="#000"; ctx.font="bold 9px Arial,sans-serif"; ctx.textAlign="left";
ctx.fillText("SYMBOL LEGEND — ASME Y32.2.3",lx+6,ly+14);
ctx.lineWidth=0.5; ctx.beginPath();ctx.moveTo(lx,ly+18);ctx.lineTo(lx+lw,ly+18);c
isoLgnd.forEach((it,i)=>{
const iy=ly+31+i*14;
ctx.save();ctx.fillStyle="#000";ctx.strokeStyle="#000";ctx.lineWidth=1.2;ctx.font
it.d(lx+4,iy);ctx.restore();
ctx.fillStyle="#111"; ctx.font="8px Arial,sans-serif"; ctx.textAlign="left"; ctx.
});
northArrow(ctx,R-56,T+56);
titleBlock(ctx,W,H,inp,job,isR?"ISOMETRIC VIEW — RISER ELEVATION":"ISOMETRIC VIEW
}
const CAT_COLOR={PIPE:"#c8860a",FITTINGS:"#3b82f6",VALVES:"#ef4444",INST:"#22c55e

export default function PipeAI(){
  const [inp,setInp]=useState({jobDesc:"",pipeSize:'2"',material:"Schedule 80 Ste
  const [job,setJob]=useState(null);
  const [tab,setTab]=useState("plan");
  const [busy,setBusy]=useState(false);
  const [fs,setFs]=useState(false);
  const [sideOpen,setSideOpen]=useState(true);
  const planRef=useRef(null);
  const isoRef=useRef(null);
  const set=k=>e=>setInp(p=>({...p,[k]:e.target.value}));

  useEffect(()=>{
```

```
  if(!job)return;
  const t=setTimeout(()=>{
    if(tab==="plan"&&planRef.current)renderPlan(planRef.current,job,inp);
    if(tab==="isometric"&&isoRef.current)renderIso(isoRef.current,job,inp);
  },30);
  return ()=>clearTimeout(t);
},[tab,job]);

function generate(){
  setBusy(true);
  setTimeout(()=>{
    const j=buildJob(inp);setJob(j);setTab("plan");
    setTimeout(()=>{if(planRef.current)renderPlan(planRef.current,j,inp);setBus
  },50);
}

function savePNG(){
  const ref=tab==="plan"?planRef.current:isoRef.current;if(!ref||!job)return;
  const a=document.createElement("a");
  a.download=job.dwgNo+"_"+(tab==="plan"?"Plan":"ISO")+".png";
  a.href=ref.toDataURL("image/png");a.click();
}

const SZ=['1/2"','3/4"','1"','1-1/4"','1-1/2"','2"','2-1/2"','3"','4"','6"','8"
const MT=["Type L Copper","Type K Copper","Sch 40 Steel","Schedule 80 Steel","3
const SY=["Domestic Hot Water","Domestic Cold Water","Steam","Condensate Return
const BT=["Commercial Office","Hospital / Healthcare","Hotel","Industrial","Hig
const JN=["Threaded","Welded (Butt)","Welded (Socket)","Grooved / Victaulic","S
const TABS=[{id:"plan",icon:"⊡",label:"Plan View"},{id:"isometric",icon:"◈",lab
const IS={width:"100%",background:"rgba(255,255,255,0.05)",border:"1px solid rg
const LS={display:"block",fontSize:"10px",fontWeight:"600",letterSpacing:"0.12e

return (
  <div style={{background:"#0a0d14",color:"#e2e8f0",height:"100vh",display:"fle
    <style>{`*{box-sizing:border-box}select option{background:#1a2035}::-webkit

    {fs&&(tab==="plan"||tab==="isometric")&&(
      <div className="fs-overlay">
        <div style={{height:"44px",background:"#050811",display:"flex",alignIte
          <div style={{display:"flex",alignItems:"center",gap:"10px"}}>
            <div style={{width:"28px",height:"28px",borderRadius:"6px",backgrou
            <span style={{fontWeight:"700",fontSize:"13px",letterSpacing:"0.1em
            {job&&<span style={{fontSize:"10px",color:"rgba(200,134,10,0.7)",ma
          </div>
          <div style={{display:"flex",gap:"8px"}}>
            <button onClick={savePNG} style={{padding:"5px 14px",background:"rg
            <button onClick={()=>setFs(false)} style={{padding:"5px 14px",backg
```

```
      </div>
    </div>
    <div style={{flex:"1",overflow:"auto",background:"#111",display:"flex",
      {tab==="plan"&&<canvas ref={planRef} width={1500} height={960} style=
      {tab==="isometric"&&<canvas ref={isoRef} width={1500} height={1100} s
    </div>
  </div>
)}

<header style={{height:"54px",background:"#050811",borderBottom:"1px solid
  <div style={{display:"flex",alignItems:"center",gap:"12px"}}>
    <div style={{width:"34px",height:"34px",borderRadius:"8px",background:"
    <div>
      <div style={{fontWeight:"700",fontSize:"15px",letterSpacing:"0.08em",
      <div style={{fontSize:"9px",color:"rgba(200,134,10,0.75)",letterSpaci
    </div>
  </div>
  <div style={{display:"flex",alignItems:"center",gap:"10px"}}>
    {job&&<div style={{display:"flex",gap:"6px"}}>
      {[{l:"DWG",v:job.dwgNo},{l:"TEST",v:job.testPSI+" PSI"},{l:"LABOR",v:
        <div key={s.l} style={{background:"rgba(255,255,255,0.05)",border:"
          <span style={{fontSize:"9px",color:"rgba(148,163,184,0.5)",letter
          <span style={{fontSize:"10px",fontWeight:"600"}}>{s.v}</span>
        </div>
      ))}
    </div>}
    <div style={{background:"rgba(200,134,10,0.1)",border:"1px solid rgba(2
      <span style={{fontSize:"10px",color:"#f0a832",fontWeight:"700"}}>v9.0
      <span style={{fontSize:"9px",color:"rgba(200,134,10,0.45)"}}>ME STAND
    </div>
  </div>
</header>

<div style={{flex:"1",display:"flex",overflow:"hidden"}}>
  <aside style={{width:sideOpen?"268px":"0px",flexShrink:"0",background:"#0
    <div style={{padding:"14px 18px 10px",borderBottom:"1px solid rgba(255,
      <div style={{fontSize:"10px",fontWeight:"700",letterSpacing:"0.22em",
    </div>
    <div style={{flex:"1",overflowY:"auto",padding:"14px 18px"}}>
      <div style={{marginBottom:"14px"}}>
        <label style={LS}>Job Description</label>
        <textarea rows={3} value={inp.jobDesc} onChange={set("jobDesc")} pl
        <div style={{fontSize:"9px",color:"rgba(148,163,184,0.3)",marginTop
      </div>
      {[
        [["Pipe Size",SZ,"pipeSize"],["Material",MT,"material"]],
        [["System Type",SY,"systemType"],["Run Length (LF)",null,"runLength
```

```
        [["Building Type",BT,"buildingType"],["Joining Method",JN,"joiningM
      ].map((row,ri)=>(
        <div key={ri} style={{display:"grid",gridTemplateColumns:"1fr 1fr",
          {row.map(([label,opts,k])=>(
            <div key={k}>
              <label style={LS}>{label}</label>
              {opts
                ? <select value={inp[k]} onChange={set(k)} style={{...IS,cu
                : <input type="number" value={inp[k]} onChange={set(k)} min
              }
            </div>
          ))}
        </div>
      ))}
      <div style={{marginBottom:"14px"}}>
        <label style={LS}>Special Conditions</label>
        <input type="text" value={inp.special} onChange={set("special")} pl
      </div>
      <div style={{padding:"8px 12px",borderRadius:"8px",background:gs(inp.
        <div style={{width:"8px",height:"8px",borderRadius:"50%",background
        <span style={{fontSize:"10px",color:gs(inp.systemType).m,fontWeight
        <span style={{fontSize:"9px",color:"rgba(148,163,184,0.4)"}}>{gs(in
      </div>
    </div>
    <div style={{padding:"14px 18px",borderTop:"1px solid rgba(255,255,255,
      <button onClick={generate} disabled={busy} style={{width:"100%",paddi
        {busy
          ? <><span style={{width:"13px",height:"13px",border:"2px solid rg
          : "Generate Drawings"
        }
      </button>
      <div style={{marginTop:"8px",textAlign:"center",fontSize:"9px",color:
    </div>
  </aside>

  <main style={{flex:"1",display:"flex",flexDirection:"column",overflow:"hi
    <div style={{height:"46px",background:"rgba(5,8,17,0.98)",borderBottom:
      <button onClick={()=>setSideOpen(o=>!o)} style={{padding:"5px 8px",ba
      {TABS.map(t=>(
        <button key={t.id} onClick={()=>setTab(t.id)} style={{padding:"5px
          <span style={{fontSize:"11px"}}>{t.icon}</span>{t.label}
        </button>
      ))}
      <div style={{marginLeft:"auto",display:"flex",gap:"6px"}}>
        {job&&(tab==="plan"||tab==="isometric")&&(
          <button onClick={()=>setFs(true)} style={{padding:"5px 13px",back
            <span>&#x26F6;</span> Full Screen
```

```
        </button>
      )}
      {job&&(tab==="plan"||tab==="isometric")&&(
        <button onClick={savePNG} style={{padding:"5px 13px",background:"
          <span>&#8595;</span> Save PNG
        </button>
      )}
    </div>
  </div>

  <div style={{flex:"1",overflow:"auto",background:tab==="plan"||tab==="i
    {!job&&(
      <div style={{height:"100%",display:"flex",flexDirection:"column",al
        <svg width="70" height="70" viewBox="0 0 70 70" style={{opacity:"
        <div style={{textAlign:"center"}}>
          <div style={{fontSize:"13px",fontWeight:"600",color:"rgba(148,1
          <div style={{fontSize:"11px",color:"rgba(148,163,184,0.12)",mar
        </div>
        <div style={{display:"flex",gap:"6px",flexWrap:"wrap",justifyCont
          {["ASME Y32.2.3 Plan View","Isometric Drawing","Material Takeof
            <div key={f} style={{padding:"4px 11px",background:"rgba(255,
          ))}
        </div>
      </div>
    )}
    {job&&tab==="plan"&&<div className="fu"><canvas ref={planRef} width={
    {job&&tab==="isometric"&&<div className="fu"><canvas ref={isoRef} wid
    {job&&tab==="details"&&(
      <div className="fu" style={{padding:"30px 34px",maxWidth:"880px"}}>
        {[{label:"Routing Strategy",key:"routing",icon:">"},{label:"Code
          <div key={item.key} style={{marginBottom:"28px",paddingBottom:"
            <div style={{display:"flex",alignItems:"center",gap:"10px",ma
              <div style={{width:"26px",height:"26px",borderRadius:"7px",
              <div style={{fontSize:"10px",fontWeight:"700",letterSpacing
            </div>
            <div style={{fontSize:"13px",lineHeight:"1.9",color:"#94a3b8"
          </div>
        ))}
        <div style={{display:"grid",gridTemplateColumns:"1fr 1fr",gap:"12
          {[{label:"Estimated Labor Hours",value:job.laborHours,unit:"HRS
            <div key={card.label} style={{background:"rgba(255,255,255,0.
              <div style={{fontSize:"9px",fontWeight:"600",letterSpacing:
              <div style={{display:"flex",alignItems:"baseline",gap:"6px"
                <span style={{fontSize:"34px",fontWeight:"800",color:card
                <span style={{fontSize:"12px",color:"rgba(148,163,184,0.4
              </div>
            </div>
```

```
      ))}
    </div>
  </div>
)}
{job&&tab==="materials"&&(
  <div className="fu" style={{padding:"26px 30px",maxWidth:"980px"}}>
    <div style={{display:"flex",gap:"7px",flexWrap:"wrap",marginBotto
      {Object.entries(job.mats.filter(m=>m.q>0).reduce((acc,m)=>{acc[
        <div key={cat} style={{display:"flex",alignItems:"center",gap
          <div style={{width:"6px",height:"6px",borderRadius:"50%",ba
          <span style={{fontSize:"9px",fontWeight:"700",color:CAT_COL
          <span style={{fontSize:"9px",color:"rgba(148,163,184,0.4)"}
        </div>
      ))}
    </div>
    <div style={{borderRadius:"10px",overflow:"hidden",border:"1px so
      <table style={{width:"100%",borderCollapse:"collapse",fontSize:
        <thead>
          <tr style={{background:"rgba(255,255,255,0.04)"}}>
            {["#","Cat","Description","Qty","Unit"].map((h,hi)=>(
              <th key={h} style={{textAlign:hi>=3?"right":"left",colo
            ))}
          </tr>
        </thead>
        <tbody>
          {job.mats.filter(m=>m.q>0).map((m,i)=>(
            <tr key={i} style={{background:i%2===0?"transparent":"rgb
              <td style={{padding:"9px 13px",color:"rgba(148,163,184,
              <td style={{padding:"9px 13px"}}>
                <span style={{display:"inline-flex",alignItems:"cente
                  <span style={{width:"5px",height:"5px",borderRadius
                  <span style={{fontSize:"9px",fontWeight:"700",color
                </span>
              </td>
              <td style={{padding:"9px 13px",color:"#cbd5e1"}}>{m.i}<
              <td style={{padding:"9px 13px",textAlign:"right",color:
              <td style={{padding:"9px 13px",textAlign:"right",color:
            </tr>
          ))}
        </tbody>
      </table>
    </div>
    <div style={{marginTop:"12px",padding:"11px 15px",background:"rgb
      Quantities include 5% waste factor. Verify against stamped draw
    </div>
  </div>
)}
```

```
          </div>
        </main>
      </div>
    </div>
  );
}
```